



Could the attached file be adapted to work with this hardware?

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I have a nice custom workstation with two 16GB RTX 5080 GPUs, 32 core/64 thread AMD Threadripper CPU, and 128GB of RAM running on headless Ubuntu Server 24.04 and I'm really curious if I were to do this if there's a decent way to harmonize both GPUs, my CPU, and my RAM because I could probably run a 70B model (maybe) locally if so, which would be awesome. I'll have to play around with this. Thanks for the video!

I now have everything I need to give a comprehensive, accurate, and entertaining answer. Let me compose it.

Great news, the workstation hardware is **absolutely capable** of running this setup — and in many ways it's *dramatically* better suited for Ollama than the laptop iGPU the runbook was written for. But the adaptation is nearly a full rewrite, not a tweak. Here's the full breakdown:

The Hardware Matchup

The runbook was written for a System76 Pangolin15 laptop with an **integrated AMD Radeon 780M GPU** that AMD officially doesn't support in ROCm — so the whole thing is essentially a box of clever workarounds: faking GPU identity, borrowing RAM via GTT, and flipping a hidden Ollama flag. Think of it as hotwiring a moped to win a scooter race.

Mitch's workstation, by contrast, is a proper race car:

Feature	Runbook Target	Mitch's Rig
GPU	AMD Radeon 780M (iGPU, shared RAM)	2× RTX 5080 (16GB VRAM each = 32GB total)
GPU backend	ROCm (with workarounds)	CUDA (natively supported)
RAM	96GB DDR5	128GB
CPU	8-core AMD laptop chip	32-core Threadripper
OS	Pop!_OS 24.04	Ubuntu Server 24.04

What to Throw Away

The runbook's three major workarounds are **all irrelevant** for the RTX 5080s :

- **No HSA_OVERRIDE_GFX_VERSION needed** — that was the trick to fool ROCm into accepting an unsupported AMD iGPU. CUDA on NVIDIA requires no such magic.
- **No GTT memory allocation** — that was specifically to lend system RAM to the integrated GPU. Discrete GPUs have their own dedicated VRAM.
- **No OLLAMA_IGPU_ENABLE=1** — that flag only exists because Ollama deliberately blocks integrated GPUs by default. Discrete GPUs are recognized automatically .

The *structure* of the runbook — install deps, configure ROCm/CUDA, set up Ollama as a systemd service, verify via journalctl — is still a useful skeleton.

What to Keep (and Adapt)

Driver Installation

Instead of the AMD ROCm repo, install NVIDIA drivers. The RTX 5080 requires **driver 570+ (open kernel module version)** . The recommended approach for Ubuntu Server 24.04 is:

```
sudo apt install nvidia-open-580 # or nvidia-driver-570-open
```

The runbook's warning about *not* installing DKMS kernel modules to avoid conflicts has a direct parallel here — on Ubuntu Server you want the open kernel module builds, not conflicting packages .

Ollama Installation & systemd Config

The install step is identical :

```
curl -fsSL https://ollama.com/install.sh | sh
```

The systemd drop-in config from the runbook is reused, but with **CUDA variables instead of ROCm variables** :

```
[Service]
Environment="CUDA_VISIBLE_DEVICES=0,1"
Environment="OLLAMA_SCHED_SPREAD=1"
Environment="OLLAMA_KEEP_ALIVE=24h"
Environment="OLLAMA_HOST=0.0.0.0:11434"
```

`OLLAMA_FLASH_ATTENTION=1` from the original runbook is **still worth keeping** — it helps on bandwidth-limited setups regardless of GPU vendor.

The 70B Dream: Can It Actually Work?

Here's where it gets interesting. A Q4_K_M quantized 70B model needs approximately 40–42GB of VRAM . Two RTX 5080s give 32GB combined — not quite enough for full GPU offload at Q4_K_M, but two real options exist:

1. **More aggressive quantization (Q3)** — gets the model down to ~28–32GB, fits entirely in dual-GPU VRAM. Speed is good (full GPU, no CPU fallback), quality is slightly reduced.
2. **Hybrid GPU+CPU offload** — load as many layers as fit into VRAM, spill the rest to the 128GB RAM. The Threadripper's 64 threads + 128GB is *genuinely excellent* for this fallback, unlike a laptop .

Ollama handles multi-GPU automatically: if the model doesn't fit on one GPU, it splits across both . Set OLLAMA_SCHED_SPREAD=1 to force an even split between the two cards .

Expected performance reality check: with both GPUs fully loaded, expect 20–40 tok/s on a 70B Q3 — usable, snappy for chat. The runbook's laptop could barely do 6 tok/s on a 14B model. This is a completely different tier .

Key Differences Summary

Runbook Step	Mitch's Adaptation
AMD ROCm repo setup	NVIDIA driver 570+ open modules
HSA_OVERRIDE_GFX_VERSION	Delete entirely
GTT kernel parameters	Delete entirely
OLLAMA_IGPU_ENABLE=1	Delete entirely
CUDA_VISIBLE_DEVICES	Add =0 , 1
OLLAMA_SCHED_SPREAD=1	Add for dual-GPU spreading
Verify with radeontop	Replace with nvidia-smi or nvtop
Verify with rocminfo	Replace with nvidia-smi

The monitoring section from the runbook is directly portable: the Open WebUI Docker command is copy-paste identical. The health-check philosophy — "run this script when it breaks" — translates 1:1, just swapping rocminfo for nvidia-smi.

Bottom line: **yes, the runbook adapts cleanly**, but about 40% of it gets deleted (the workarounds), 40% stays the same (Ollama setup, systemd, Open WebUI), and 20% gets swapped (ROCm → CUDA, AMD monitoring → NVIDIA monitoring). The Threadripper + 128GB RAM combo means even if the 70B model spills off the GPUs, the fallback is fast enough to still be genuinely useful — which is more than the laptop could ever say.



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